Complexity and Recurrences

Algorithmic Performance and Mathematical Analysis

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Course Outline

1. Introduction to Complexity Analysis
2. Advanced Complexity Analysis





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Outline

1 Introduction to Complexity Analysis

2 Advanced Complexity Analysis





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Why Analyze Algorithms?

The Fundamental Question

Given multiple algorithms that solve the same problem, which one should we choose?

- **Performance:** How fast does the algorithm run?
- **Resources:** How much memory does it use?
- **Scalability:** How does it handle large inputs?
- **Predictability:** Can we predict its behavior?

Real-World Impact

- User experience depends on algorithm speed
- Server costs increase with inefficient algorithms
- Mobile apps drain battery with poor algorithms
- Scientific computing requires efficient solutions

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Elementary Operations Identification I

Definition

An **elementary operation** is a basic computation that takes constant time, regardless of input size.

Common Elementary Operations:

- Assignment: `x = 5`
- Arithmetic: $+, -, *, /$





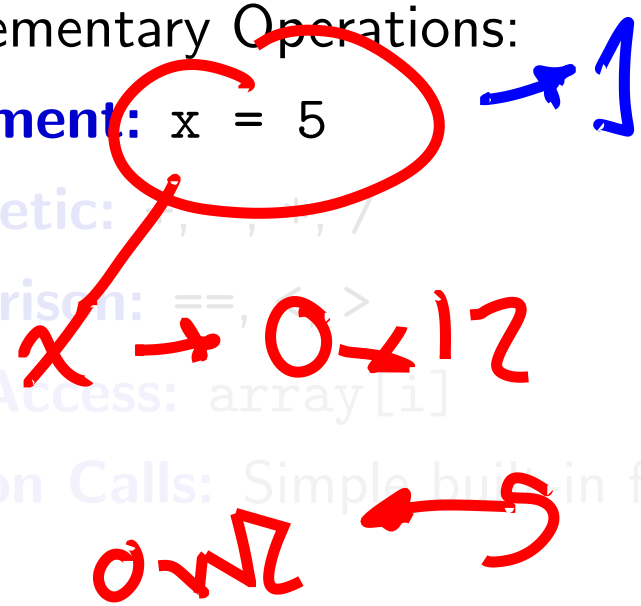
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↳ not execution





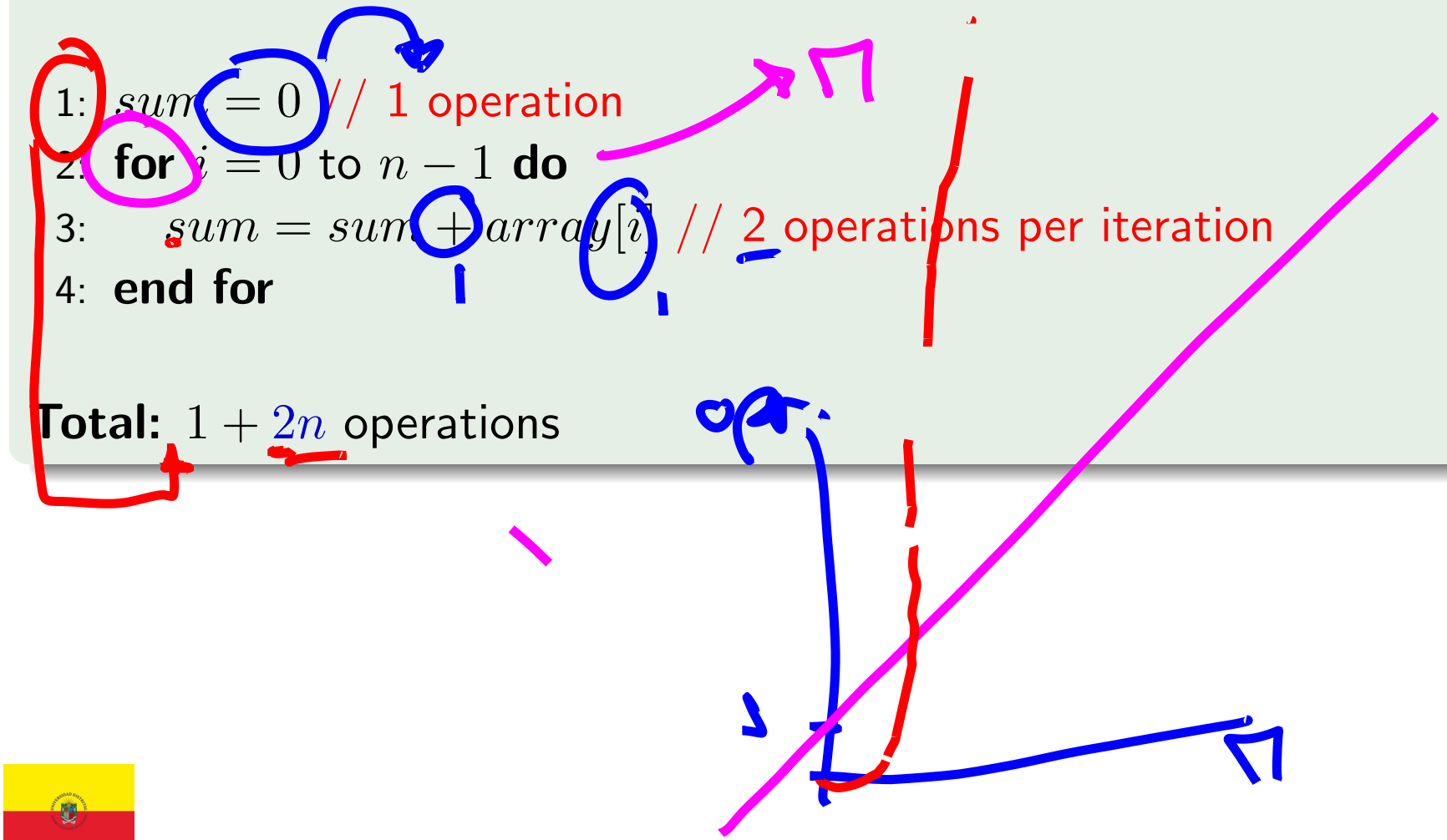
Elementary Operations Identification II

Example

Count operations in simple code:

1: $sum = 0$ // 1 operation
2: **for** $i = 0$ to $n - 1$ **do**
3: $sum = sum + array[i]$ // 2 operations per iteration
4: **end for**

Total: $1 + 2n$ operations





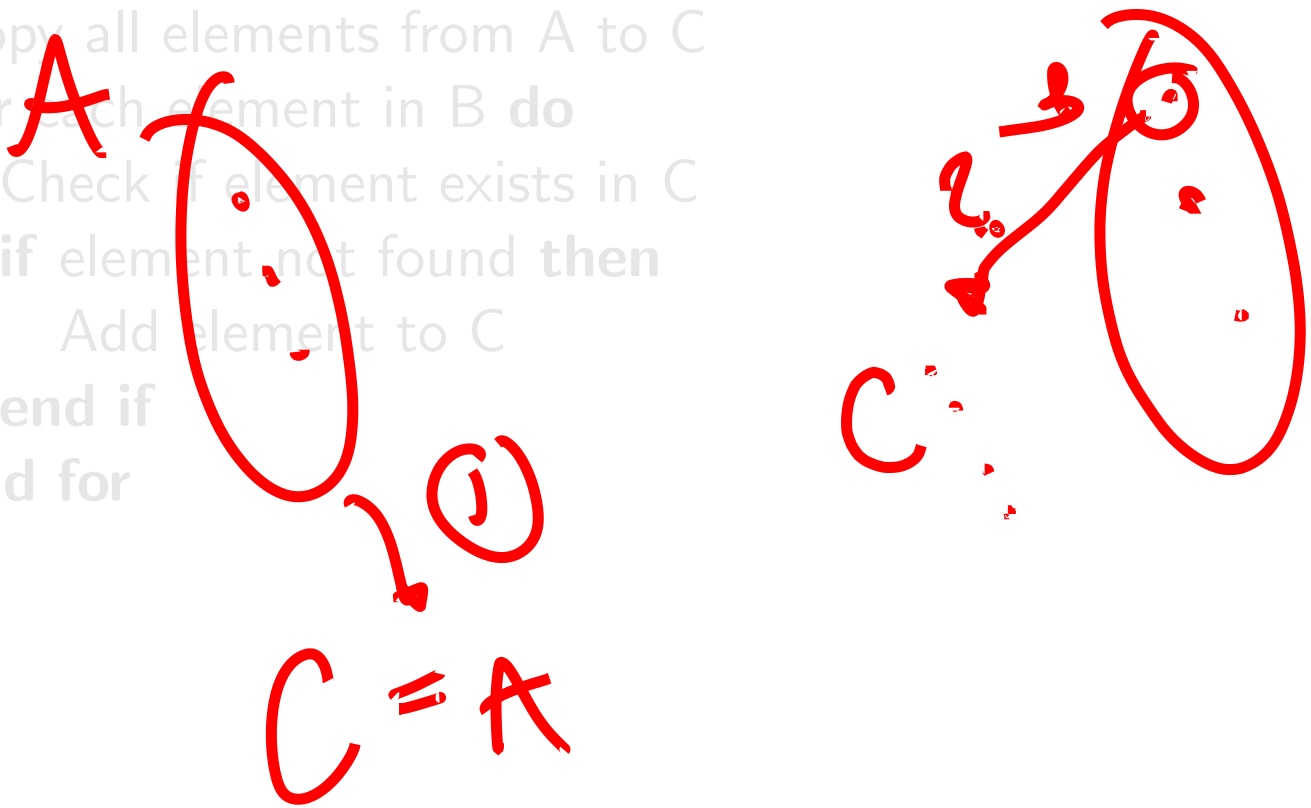


Study Case: Counting Operations in Arrays Union I

Problem

Given two arrays A and B, create array C that contains all elements from both arrays without duplicates.

```
1: Copy all elements from A to C
2: for each element in B do
3:     Check if element exists in C
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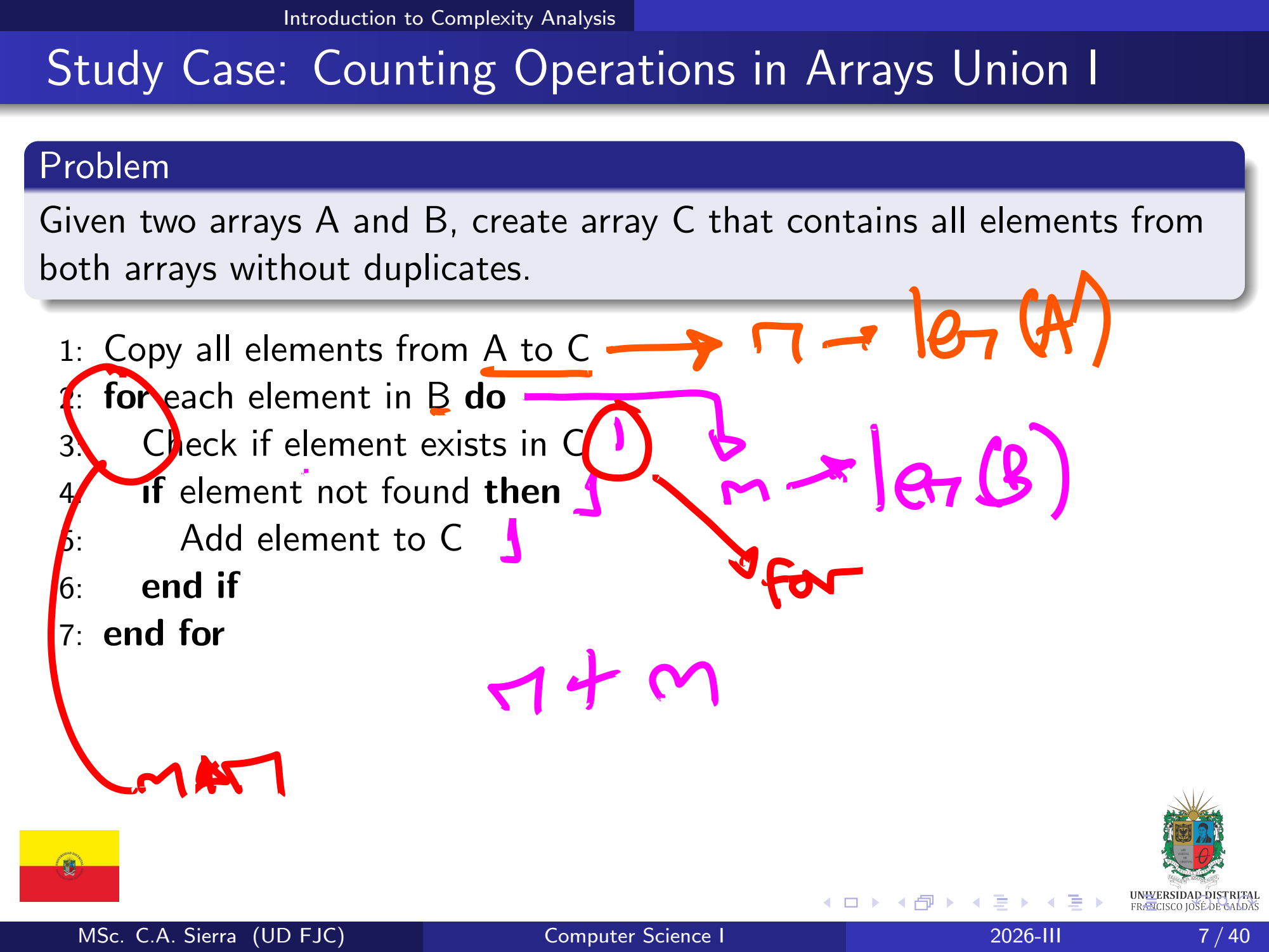


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Study Case: Counting Operations in Arrays Union II

Operation Analysis

- **Copy from A:** n operations
- **For each element in B:** m iterations
- **Search in C:** Up to n comparisons per element
- **Total worst case:** $n + m * n = n(1 + m)$ operations

Key Insight

The nested search makes this algorithm slow for large arrays! Performance grows quadratically.



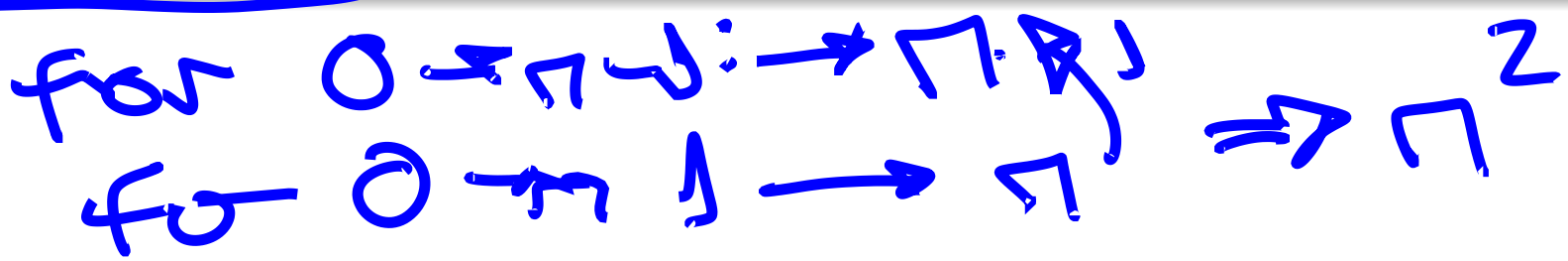
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Best, Average, Worst Case Scenarios

Three Perspectives on Algorithm Performance

Different inputs can lead to dramatically different execution times for the same algorithm.

Best Case

- Minimum operations
- Most favorable input
- Example: Finding element at first position

Average Case

- Expected performance
- Typical input
- Example: Finding element in middle

Worst Case

- Maximum operations
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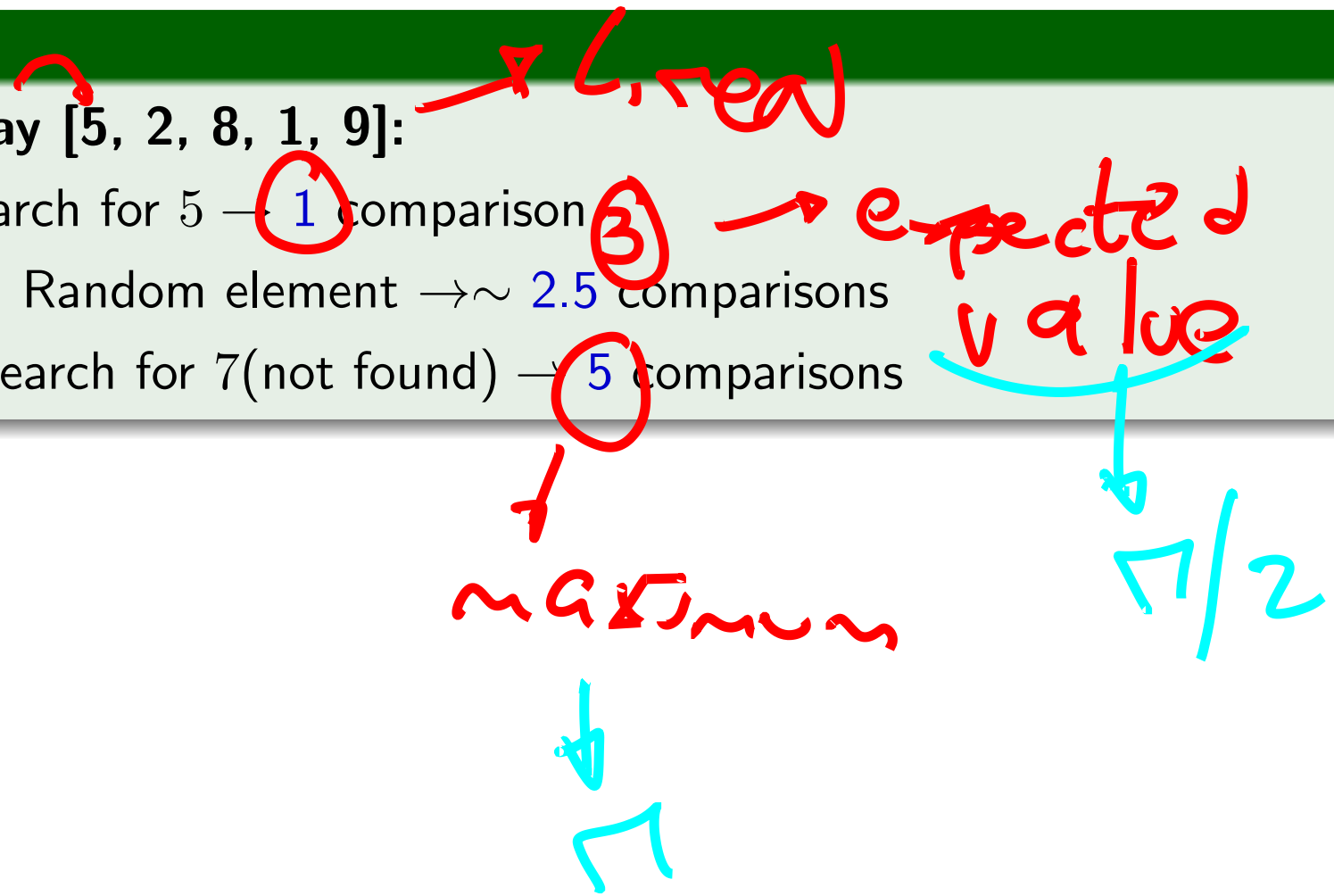


Best, Average, Worst Case Scenarios II

Example

Search in array [5, 2, 8, 1, 9]:

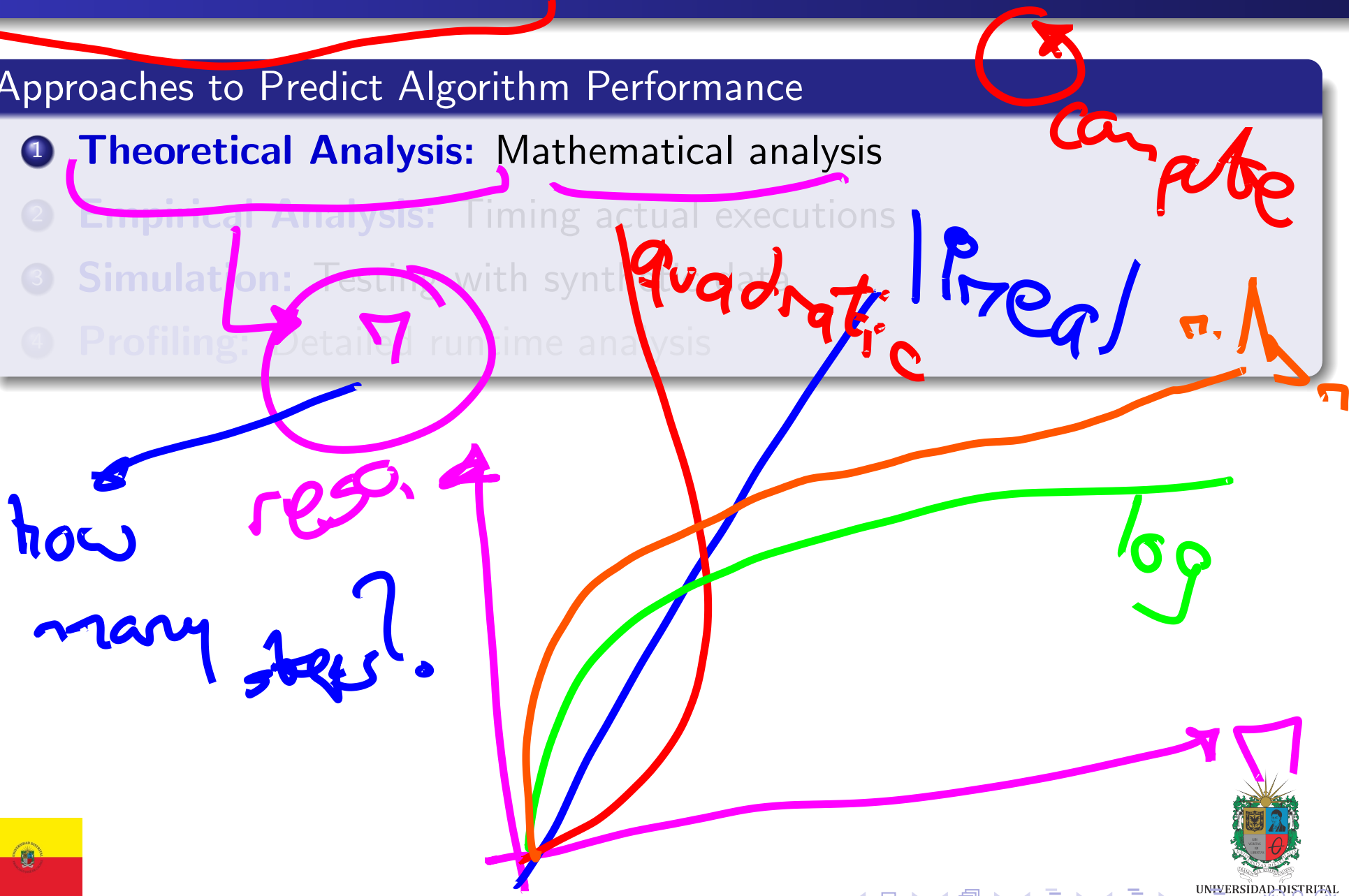
- **Best:** Search for 5 $\rightarrow$ 1 comparison
- **Average:** Random element $\rightarrow\sim$ 2.5 comparisons
- **Worst:** Search for 7(not found) $\rightarrow$ 5 comparisons



Performance Prediction Methods

Approaches to Predict Algorithm Performance

1. **Theoretical Analysis:** Mathematical analysis
2. **Empirical Analysis:** Timing actual executions
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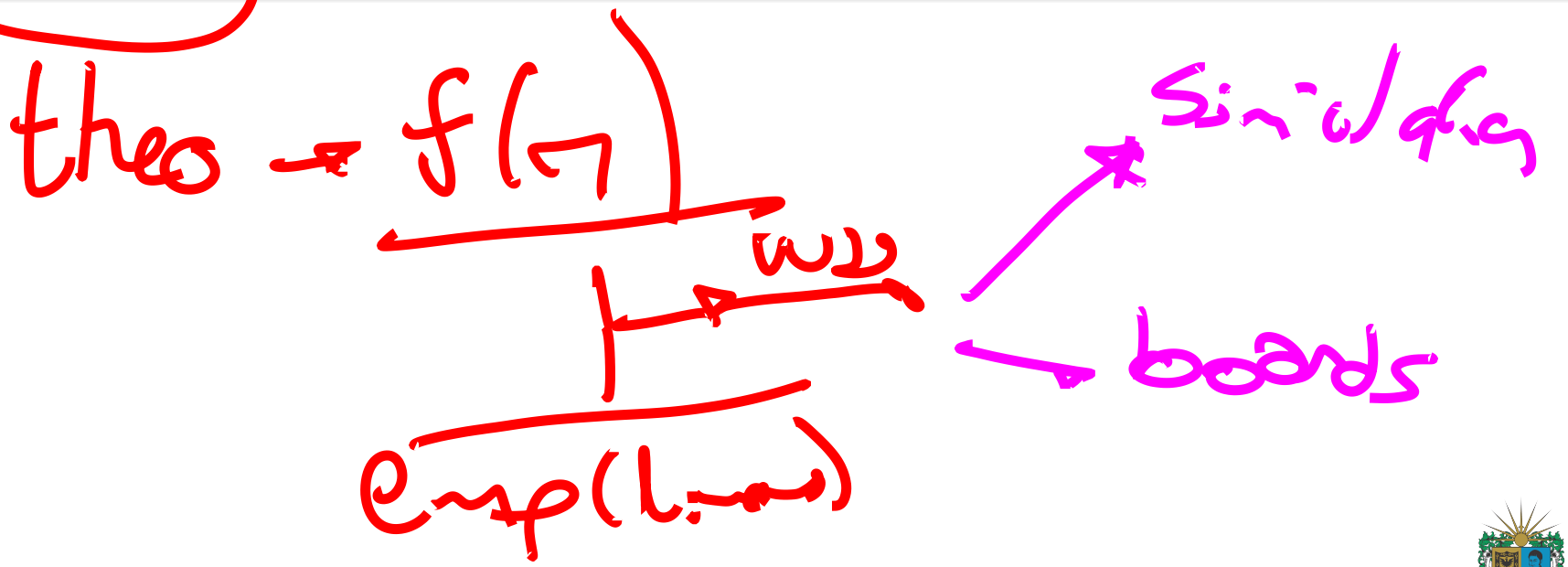




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Performance Prediction Methods II

Theoretical Analysis

Advantages:

- Machine-independent
- Covers all possible inputs
- Mathematical precision

Limitations:

- Ignores implementation details
- May not reflect real hardware

Empirical Analysis

Advantages:

- Reflects real performance
- Shows actual constants
- Platform-specific results

Limitations:

- Machine-dependent
- Limited input coverage
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Best Practice

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Time vs. Space Trade-offs I

Definition

A time-space trade-off occurs when we can reduce time by using more memory, or reduce memory at the cost of increased time.

Trade-off Decision Factors:

- Available memory constraints
- Time requirements
- Cost of computation vs. storage
- Frequency of algorithm execution

- time
+ memory
(cheap)

- memory
+ time
/ apps





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Time vs. Space Trade-offs II

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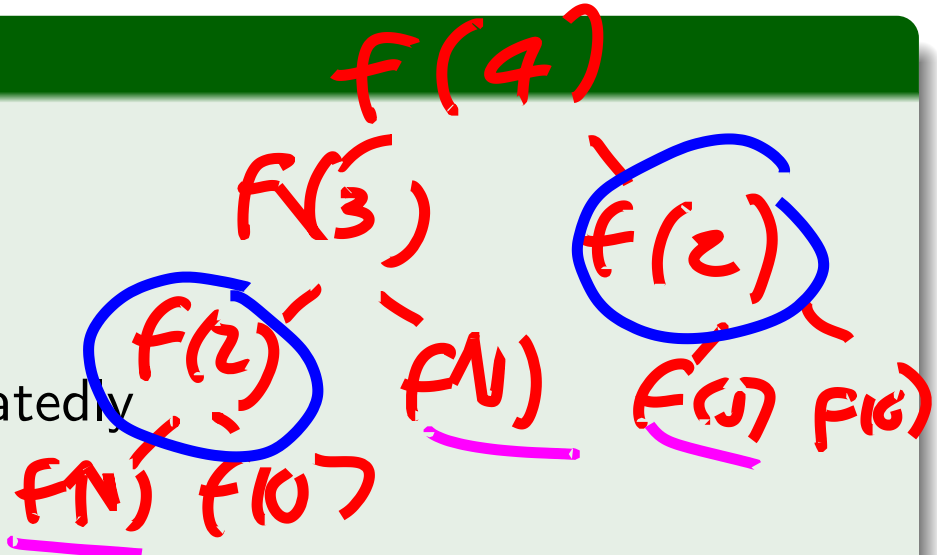
Fibonacci Calculation:

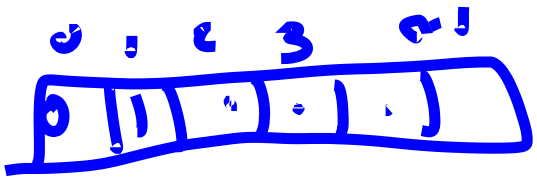
Space-Efficient (Recursive):

- Uses minimal extra memory
- But recalculates same values repeatedly
- Very slow for large numbers

Time-Efficient (Memoization):

- Stores previously calculated values
- Uses more memory
- Much faster for large numbers





for i = 2 → n-1:
a[i] = a[i-1] + a[i-2]

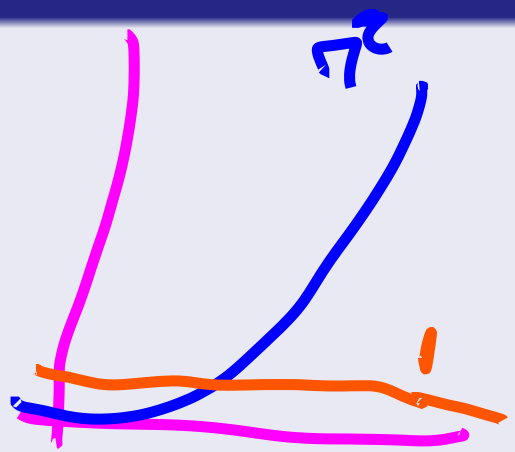




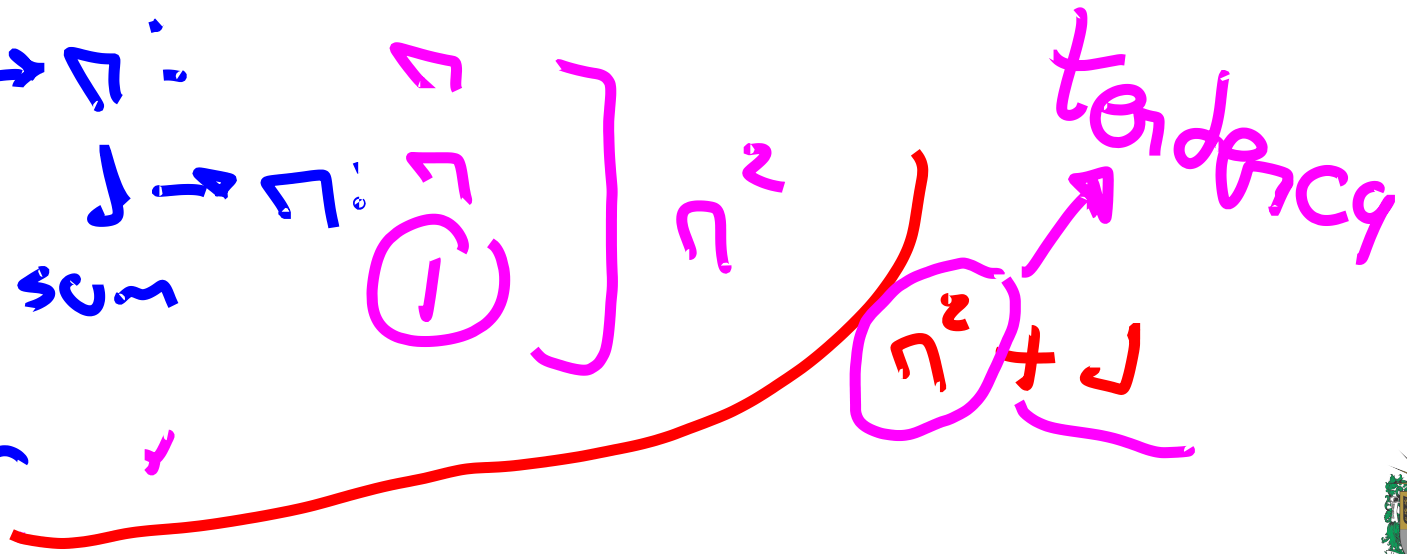
Systematic Complexity Calculation Methods I

Step-by-Step Approach

1. **Identify:** All loops and recursive calls
2. **Count:** Operations in each section
3. **Multiply:** Operations by execution frequency
4. **Sum:** All contributions
5. **Simplify:** Focus on dominant terms



for i → n:
for j → n:
sum
dimension







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Systematic Complexity Calculation Methods II

Example

Nested Loop Analysis:

```
1: for i = 0 to n − 1 do
2:     for j = 0 to n − 1 do
3:         result = array[i] + array[j]
4:     end for
5: end for
```

Operation Count:

- Outer loop runs n times
- Inner loop runs n times for each outer iteration
- Total operations: $n * n = n^2$ operations



Study Case: Complexity Calculation for `QuickSort` I

QuickSort Algorithm Structure

1: Choose a pivot element
2: Partition array around pivot
3: Recursively sort left part
4: Recursively sort right part



Partition Operation

- Compare each element with pivot
- Rearrange elements (smaller left, larger right)
- This takes n operations for n elements

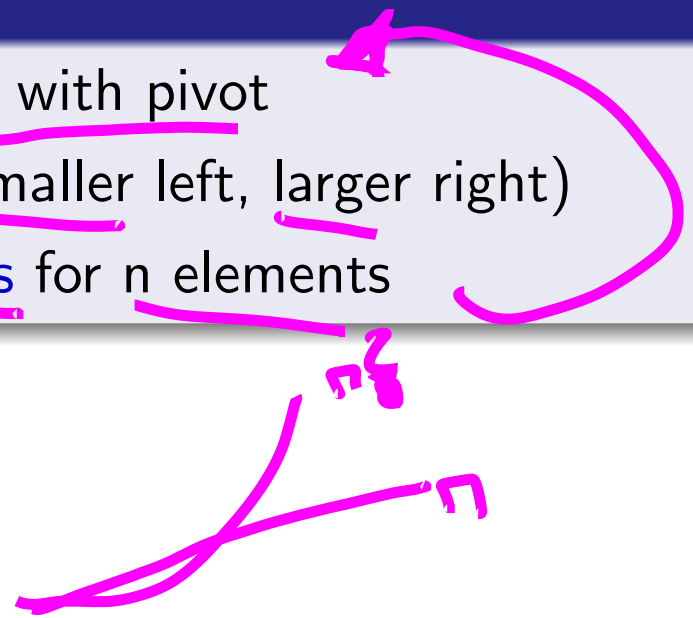
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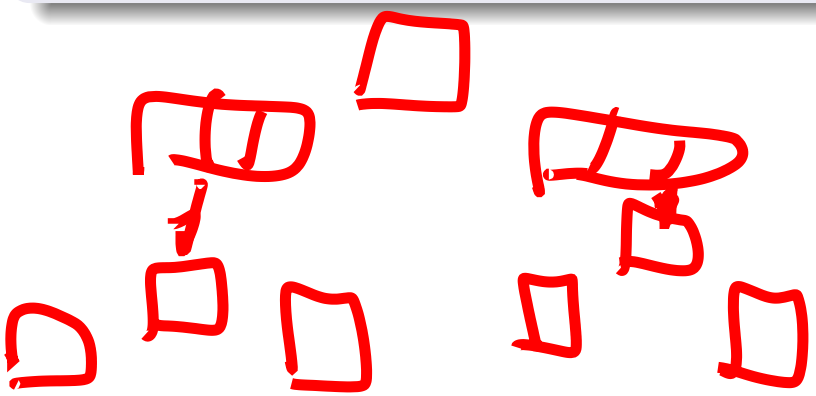




Study Case: Complexity Calculation for QuickSort II

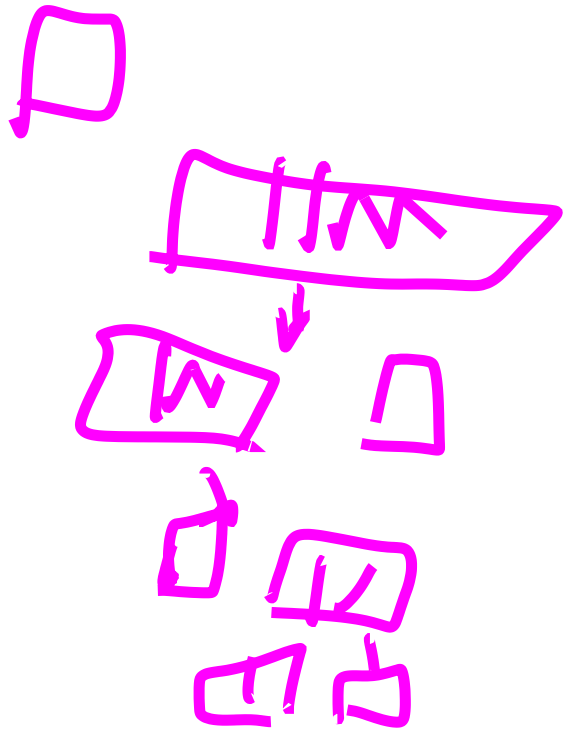
Best Case

- Pivot divides array in half
- Balanced partitions
- Fast performance



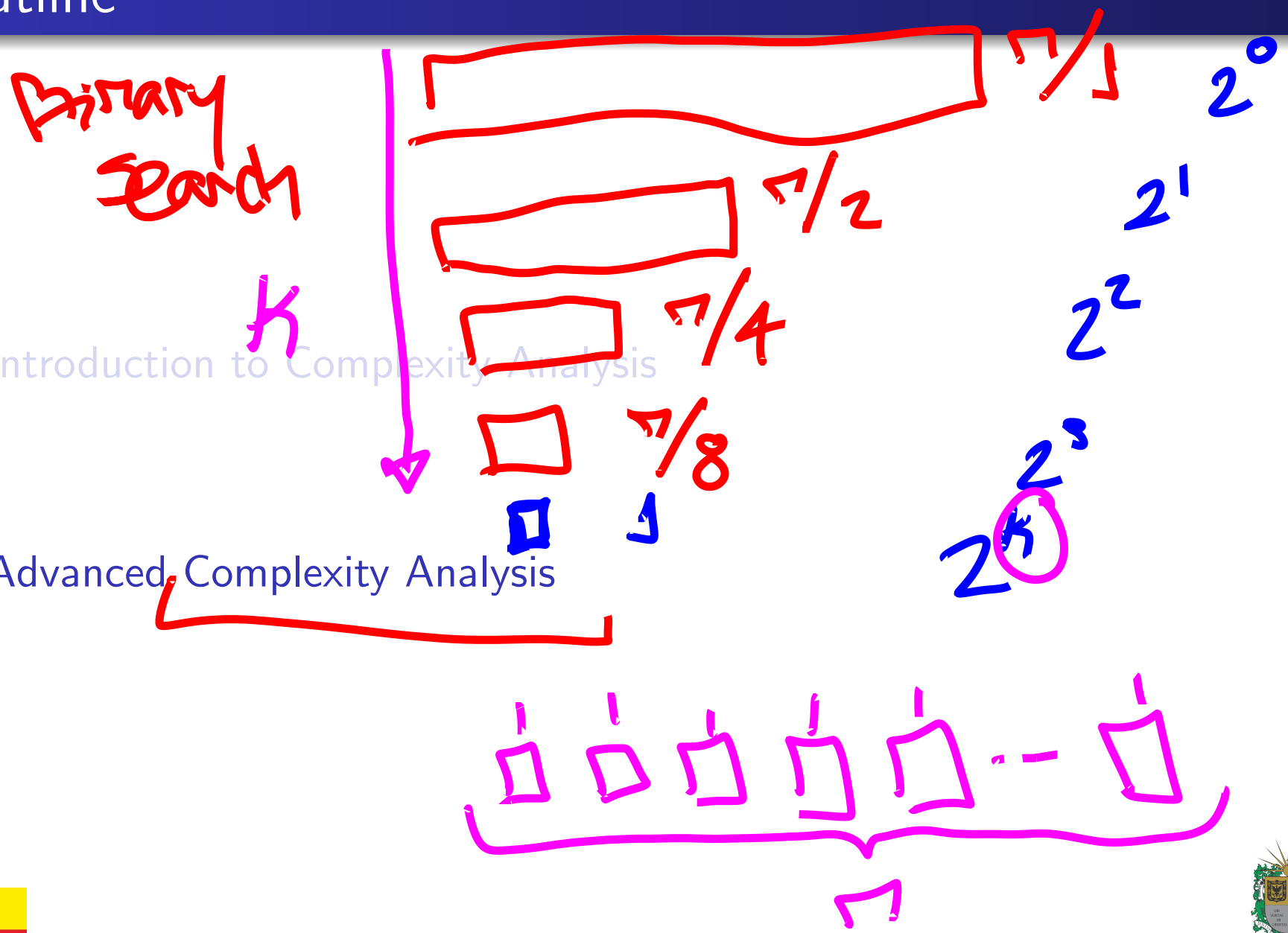
Worst Case

- Pivot is always smallest/largest
- Unbalanced partitions
- Slow performance



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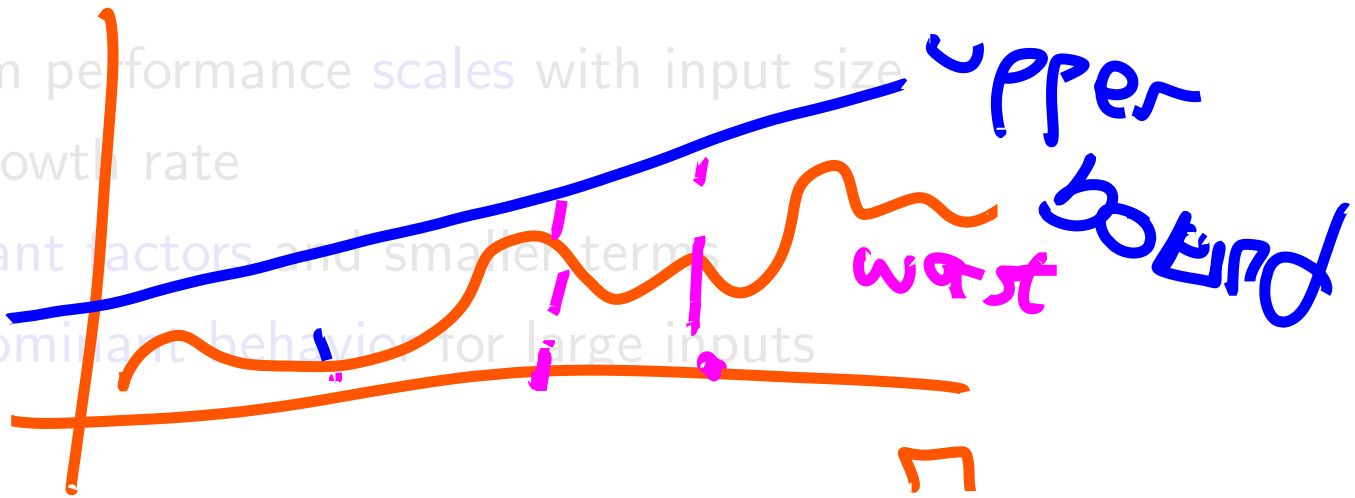
Big O Notation: Definition and Meaning

Definition

Big O notation describes the upper bound of an algorithm's growth rate for large inputs.

What Big O Tells Us:

- How algorithm performance scales with input size
- Worst-case growth rate
- Ignores constant factors and smaller terms
- Focuses on dominant behavior for large inputs







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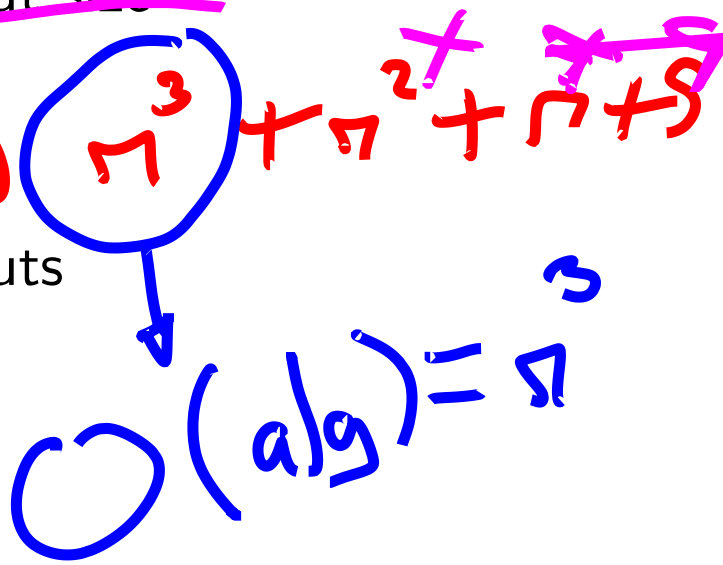
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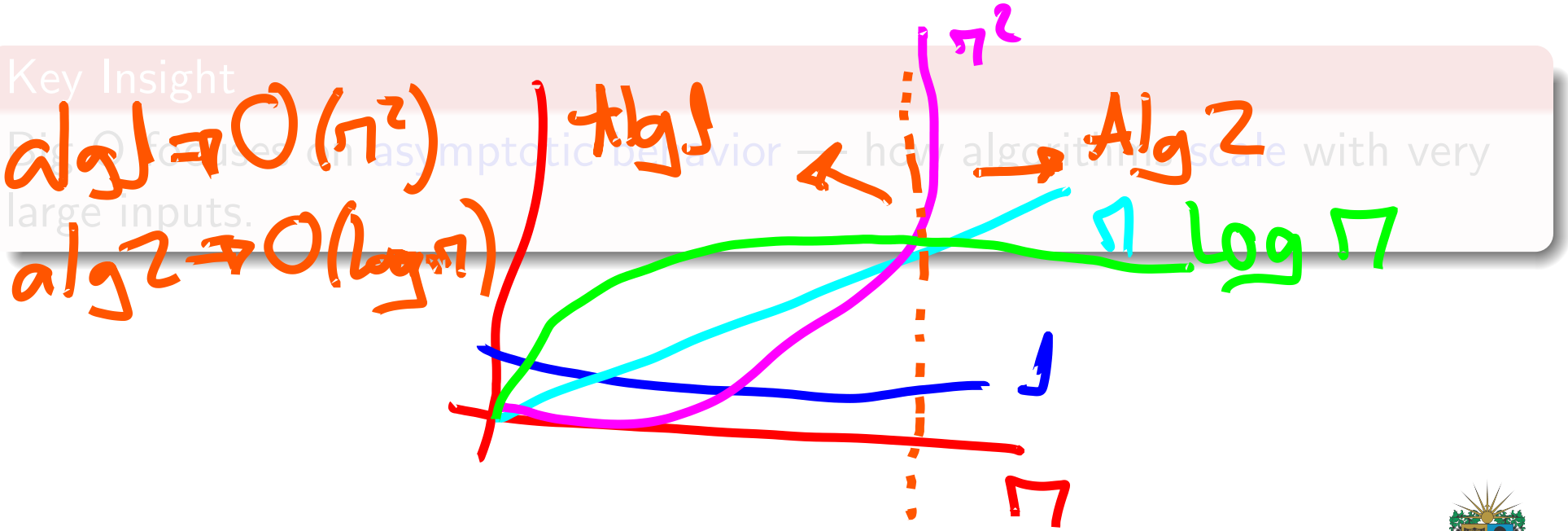


Big O Notation: Classifications

22/11/2025 → 1

Common Big O Classifications:

- $O(1)$ — Constant: Always same time
- $O(n)$ — Linear: Time grows with input size
- $O(n^2)$ — Quadratic: Time grows with square of input
- $O(logn)$ — Logarithmic: Very efficient growth



Key Insight

Big O focuses on asymptotic behavior — how algorithms scale with very large inputs.

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$f(x) = 5x^2 + 2x$

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Big O focuses on asymptotic behavior — how algorithms scale with very large inputs.

$O(n^2) \Rightarrow \exists c / \forall x \rightarrow cn^2 \geq f(x)$

$\geq \forall$ cn^2 $5x^2 + 2x$





Upper Bound Concept and Mathematical Definition

Understanding Upper Bounds

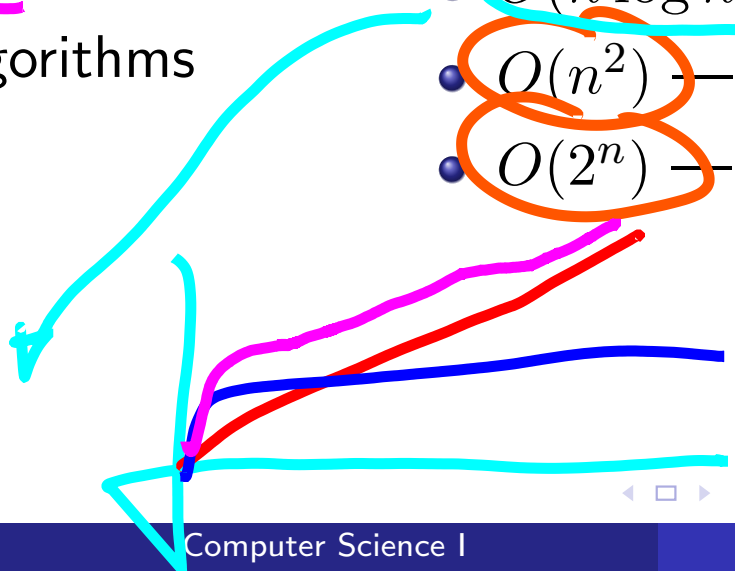
An **upper bound** guarantees that an algorithm will never perform worse than the bound.

Big O Meaning:

- Sets a ceiling on performance
- Guarantees maximum time
- Used for worst-case analysis
- Helps compare algorithms

Common Growth Orders:

- $O(1)$ — Constant
- $O(logn)$ — Logarithmic
- $O(n)$ — Linear
- $O(n \log n)$ — Linearithmic
- $O(n^2)$ — Quadratic
- $O(2^n)$ — Exponential



Study Case: First Letter in a Sentence Complexity

Problem

Find the first alphabetical letter in a given sentence string.

```
1: for each character in sentence do
2:     if character is a letter then
3:         return character
4:     end if
5: end for
6: return null
```

Best: First character is a letter

- Only check 1 character
- Constant time: $O(1)$

Worst: No letters in sentence

- Check every character
- Linear time: $O(n)$

Even though best case is $O(1)$, we report $O(n)$ because Big O represents the worst-case upper bound.

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Study Case: Linear Search Algorithm Complexity

Problem

Search for a target element in an unsorted array.

```
1: for each element in array do
2:     if element equals target then
3:         return index
4:     end if
5: end for
6: return -1
```

- **Best Case:** Element at first position, 1 comparison $\rightarrow O(1)$
- **Worst Case:** Element not in array, n comparisons $\rightarrow O(n)$
- **Average Case:** Element in middle, $n/2$ comparisons $\rightarrow O(n)$





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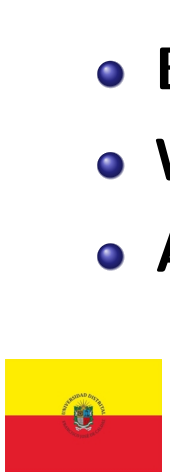
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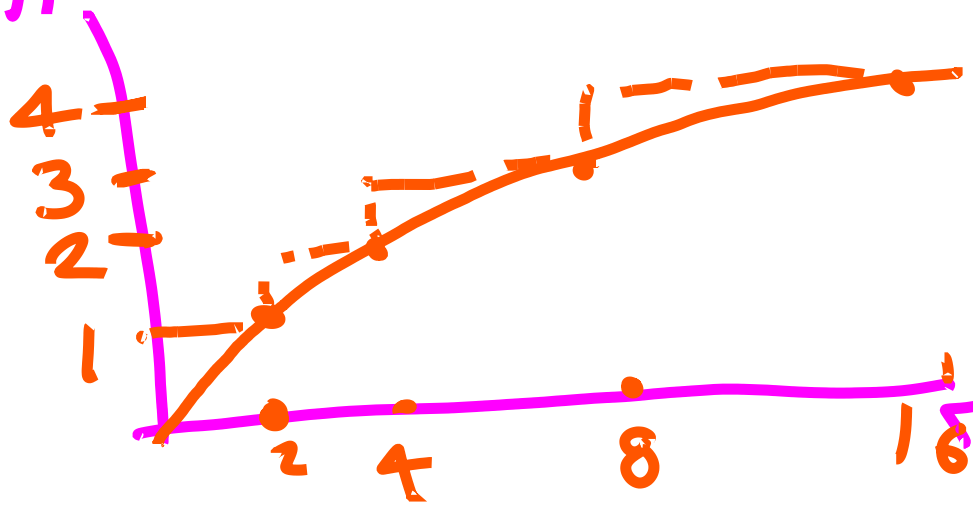


Study Case: Binary Search Algorithm Complexity I

Problem

Search for a target element in a sorted array.

```
1:  while search space not empty do
2:      Find middle element
3:      if middle equals target then
4:          return index
5:      else if middle ¡ target then
6:          search right half
7:      else
8:          search left half
9:      end if
10: end while
11: return -1
```





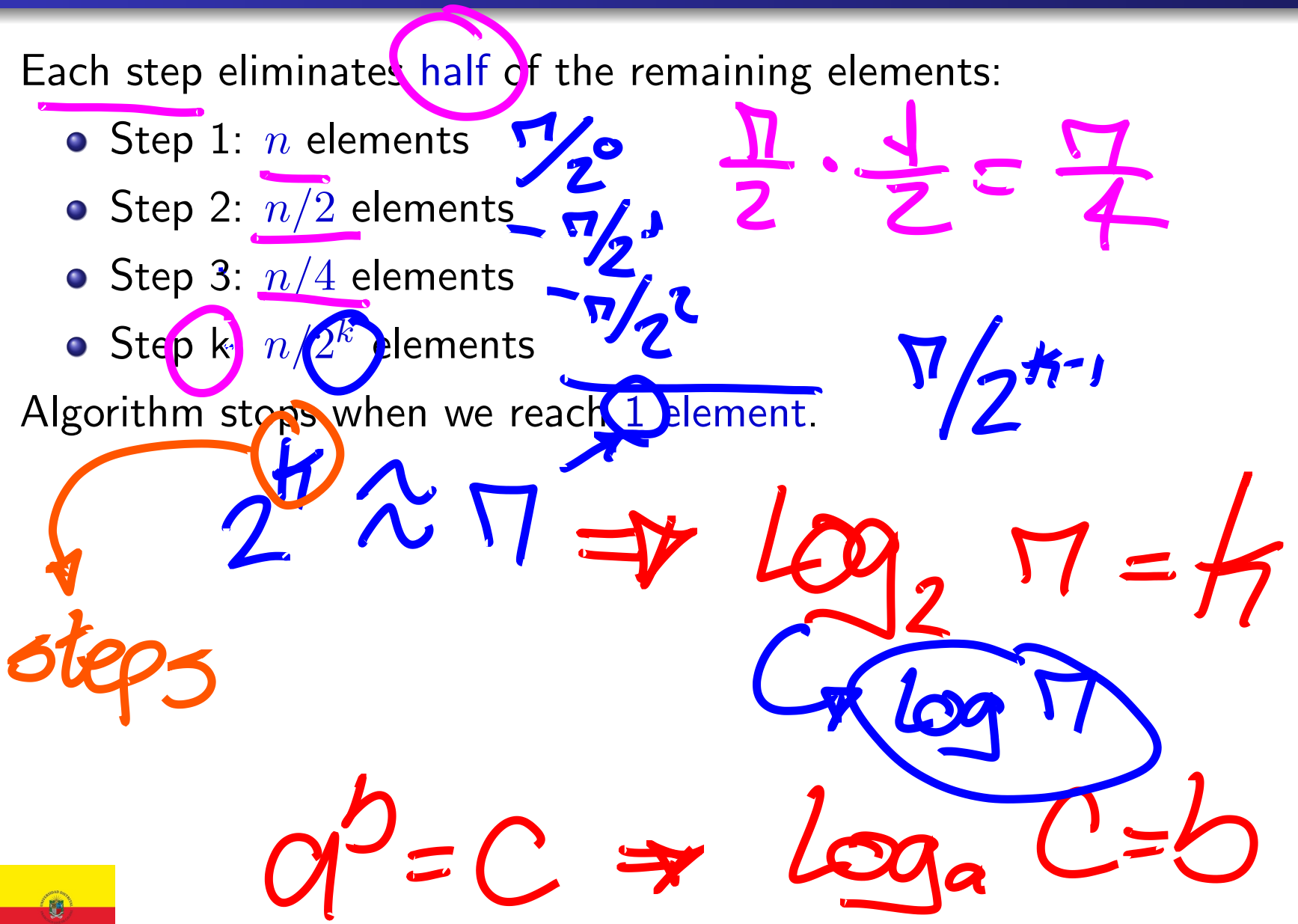


Study Case: Binary Search Algorithm Complexity II

Each step eliminates half of the remaining elements:

- Step 1: n elements
- Step 2: $n/2$ elements
- Step 3: $n/4$ elements
- Step k: $n/2^k$ elements

Algorithm stops when we reach 1 element.



Big Omega (Ω) and Big Theta (Θ) Notations I

Big Omega (Ω)

Lower Bound:

- Minimum time required
- Best-case analysis
- "At least this fast"
- Guarantees algorithm won't be faster

Big Theta (Θ)

Tight Bound:

- Exact growth rate
- Best and worst are same
- "Exactly this fast"
- Most precise description



Big Omega (Ω) and Big Theta (Θ) Notations II

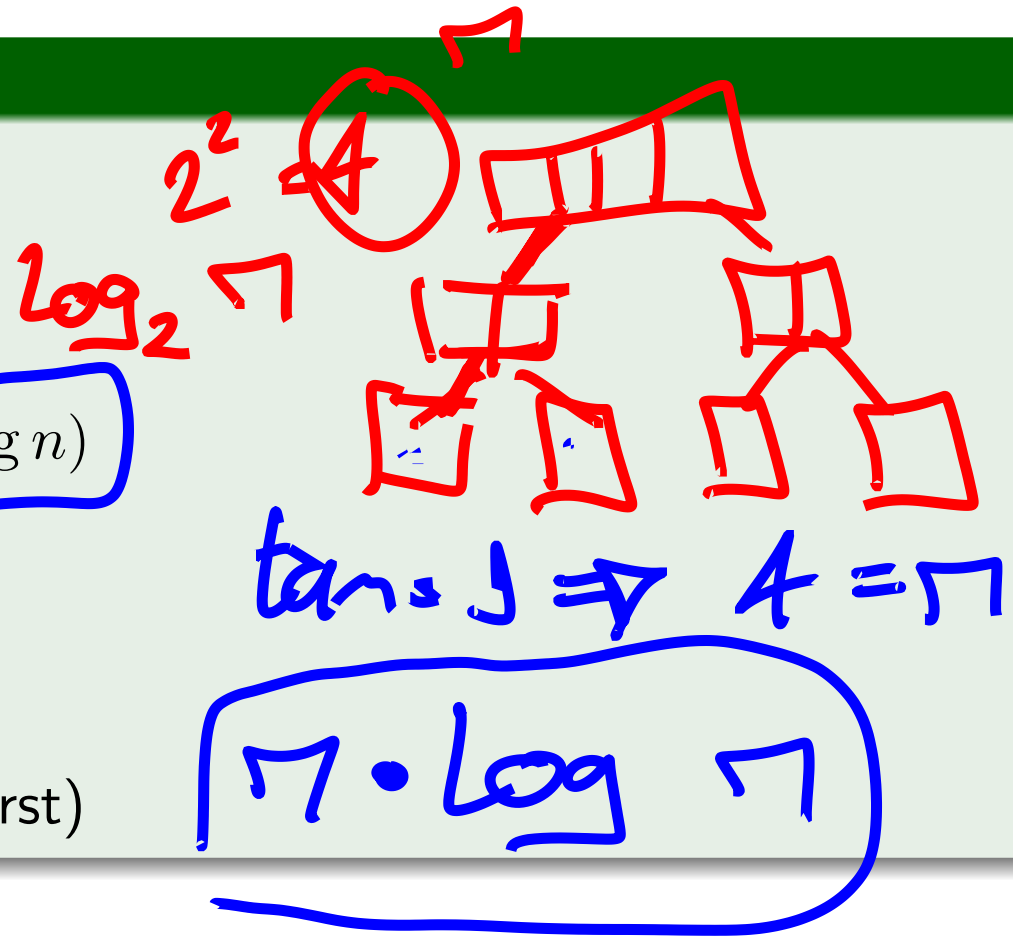
Example

Merge Sort:

- Best case: $\Omega(n \log n)$
- Worst case: $O(n \log n)$
- Since best $=$ worst: $\Theta(n \log n)$

Linear Search:

- Best case: $\Omega(1)$
- Worst case: $O(n)$
- No tight bound (best $\neq$ worst)

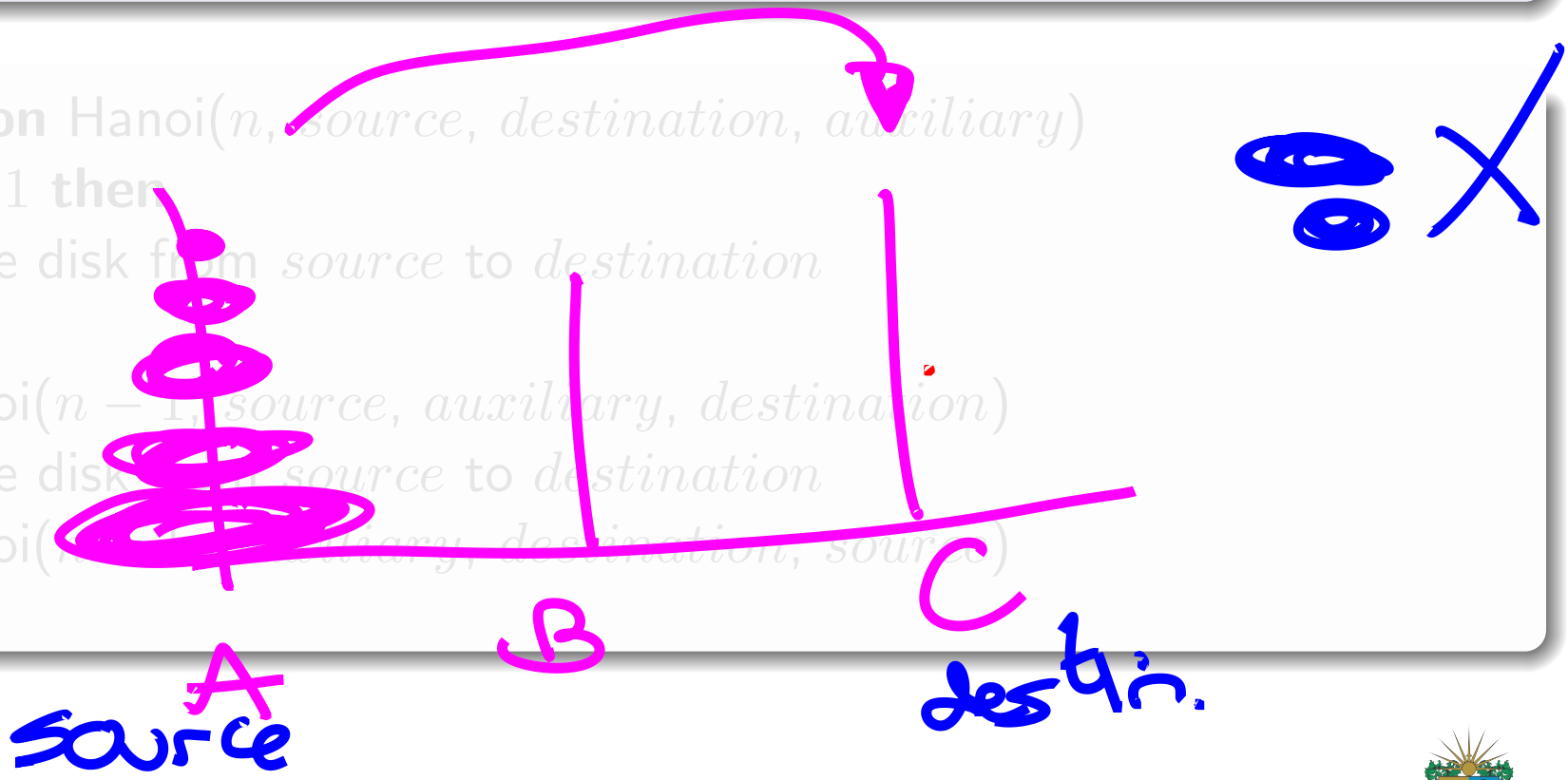


Study Case: Tower of Hanoi Problem I

Problem

Move n disks from source rod to destination rod using auxiliary rod. Rules: only move one disk at a time, never place larger disk on smaller one.

```
1: function Hanoi(n, source, destination, auxiliary)
2: if n = 1 then
3:     move disk from source to destination
4: else
5:     Hanoi(n − 1, source, auxiliary, destination)
6:     move disk from source to destination
7:     Hanoi(n − 1, auxiliary, destination, source)
8: end if
```



Study Case: Tower of Hanoi Problem I

Problem

Move n disks from source rod to destination rod using auxiliary rod. Rules: only move one disk at a time, never place larger disk on smaller one.

1: **function** Hanoi(n, $source$, $destination$, $auxiliary$)
2: **if** $n = 1$ **then**
3: move disk from $source$ to $destination$
4: **else**
5: Hanoi($n - 1$, $source$, $auxiliary$, $destination$)
6: move disk from $source$ to $destination$
7: Hanoi($n - 1$, $auxiliary$, $destination$, $source$)
8: **end if**





Study Case: Tower of Hanoi Problem II

Recurrence Analysis

- $T(1) = 1$ (base case: move one disk)
- $T(n) = 2T(n-1) + 1$ (two recursive calls + one move)

Solution:

- $T(n) = 2^n - 1$
- Complexity: $O(2^n)$ — exponential!

Real-World Implication

For 64 disks: $2^{64} - 1 \approx 18$ billion billion moves!

Study Case: Tower of Hanoi Problem II

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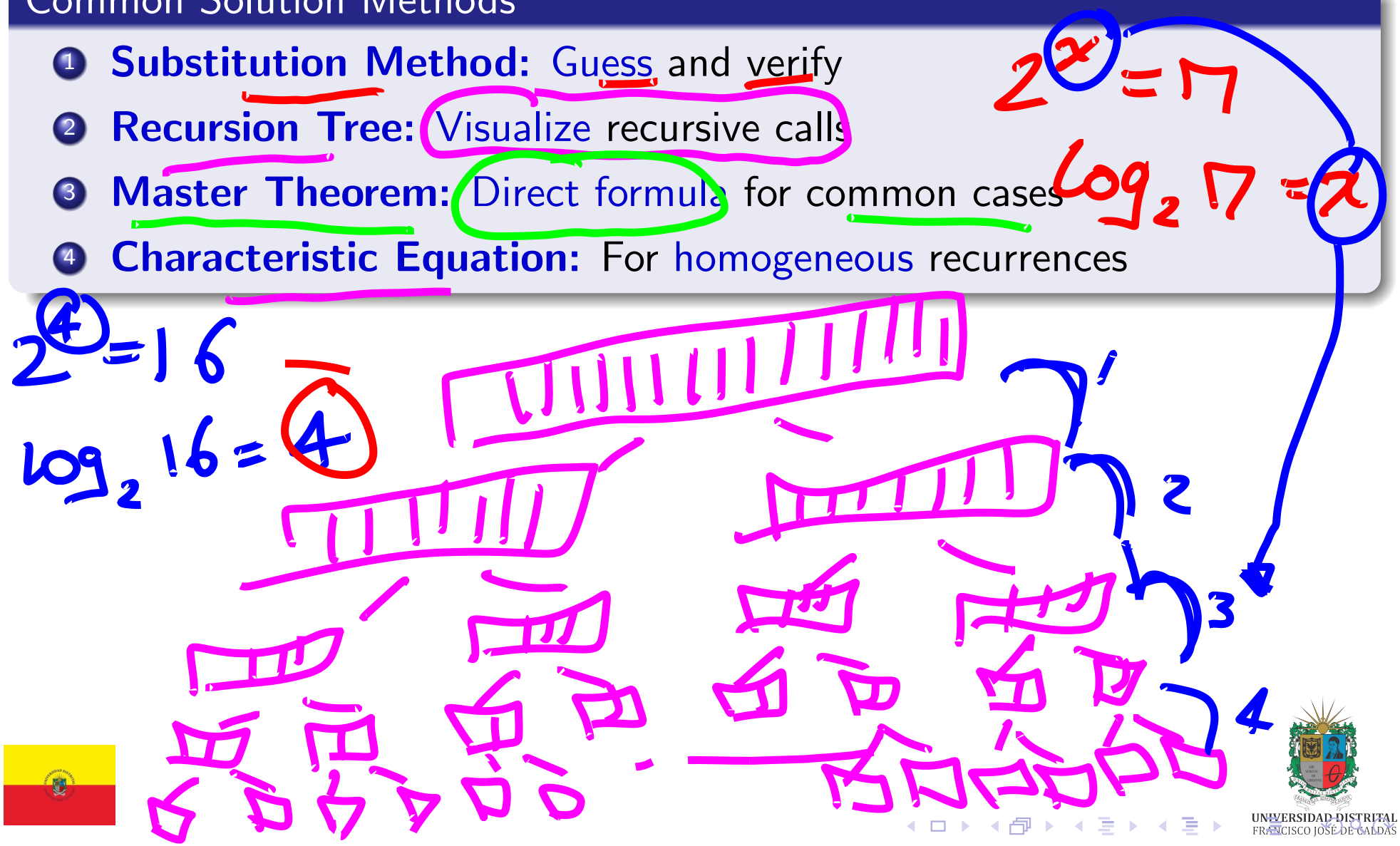




Solving Recurrence Relations

Common Solution Methods

1. **Substitution Method:** Guess and verify
2. **Recursion Tree:** Visualize recursive calls
3. **Master Theorem:** Direct formula for common cases
4. **Characteristic Equation:** For homogeneous recurrences



Solving Recurrence Relations

Example

Recursion Tree Method for $T(n) = 2T(n/2) + n$:

- Level 0: n work
- Level 1: $2 \times (n/2) = n$ work
- Level 2: $4 \times (n/4) = n$ work
- ...
- Total levels: $\log n$
- Total work: $n \times \log n = O(n \log n)$



Method Selection

- Simple cases: Direct solving
- Divide-and-conquer: Master theorem
- Complex cases: Substitution method

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Method Selection

- Simple cases: Direct solving
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- Complex cases: Substitution method

Master Theorem Introduction I

Master Theorem

For recurrences of the form $T(n) = aT(n/b) + f(n)$:

Case 1: $f(n)$ grows slower than $n^{(\log_b a)}$

- $T(n) = O(n^{(\log_b a)})$

Case 2: $f(n)$ grows at same rate

- $T(n) = O(n^{(\log_b a)} \log n)$

Case 3: $f(n)$ grows faster

- $T(n) = O(f(n))$



Master Theorem Introduction II

Examples

Merge Sort:

- $T(n) = 2T(n/2) + n$
- $a = 2,\ b = 2,\ f(n) = n$
- Case 2: $O(n \log n)$

Binary Search:

- $T(n) = T(n/2) + 1$
- $a = 1,\ b = 2,\ f(n) = 1$
- Case 2: $O(\log n)$

When to Use

Master theorem works for many divide-and-conquer algorithms but has limitations.





Master Theorem Introduction II

Examples

Merge Sort:

- $T(n) = 2T(n/2) + n$
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- $a = 1,\ b = 2,\ f(n) = 1$
- Case 2: $O(\log n)$

When to Use

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Recursive Algorithm Complexity Calculation I

Systematic Approach

1. **Identify:** Base case(s) and recursive case(s)
2. **Set up:** Recurrence relation
3. **Solve:** Using appropriate method
4. **Verify:** Check solution makes sense





Recursive Algorithm Complexity Calculation II

Merge Sort Analysis

Step 1: Identify cases

- Base: Array size $1 \rightarrow O(1)$
- Recursive: Divide, sort halves, merge

Step 2: Set up recurrence

- $T(n) = 2T(n/2) + O(n)$

Step 3: Solve using Master Theorem

- Case 2: $T(n) = O(n \log n)$

Step 4: Verify

- Makes sense: logarithmic levels, linear work per level





Summary I

Key Takeaways

- **Complexity Analysis:** Essential for algorithm comparison and selection
- **Big O Notation:** Describes growth rate for large inputs
- **Recursion:** Powerful technique requiring careful base case design
- **Recurrence Relations:** Mathematical tools for recursive algorithm analysis





Summary II

Practical Skills Gained

- Count operations systematically
- Classify algorithms by growth rate
- Design recursive solutions correctly
- Analyze recursive algorithm complexity
- Apply Master Theorem to common patterns

Next Steps

- Practice complexity analysis on various algorithms
- Implement recursive solutions with proper optimization
- Study advanced recurrence solving techniques
- Apply knowledge to real-world algorithm selection





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Outline

1 Introduction to Complexity Analysis

2 Advanced Complexity Analysis





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Thanks!

Questions?



Repo: *https://github.com/EngAndres/ud-public/tree/main/courses/computer-sciences-i*